

WEEK 6 – HEALTHCARE ANALYSIS DASHBOARD REPORT

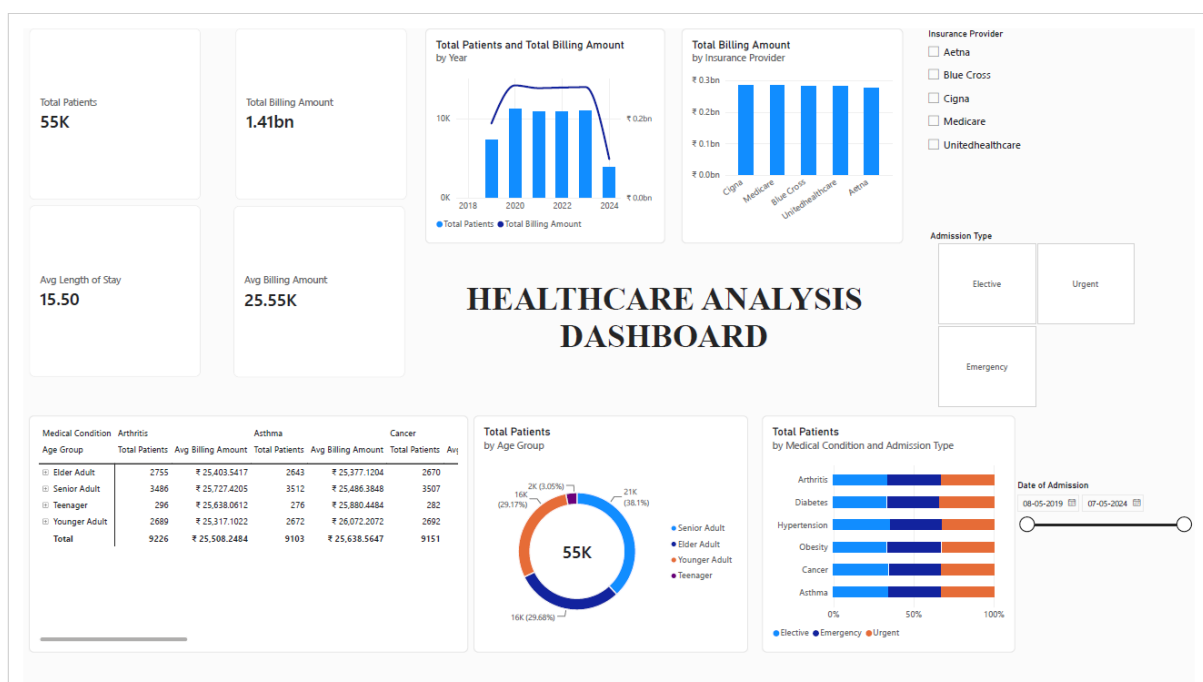
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B.SC DS & AI

1. Introduction

This project presents a **Healthcare Analysis Dashboard** created using Power BI. The dashboard provides a clear view of patient information, billing amounts, admission types, medical conditions, age groups, and insurance providers. It helps in understanding healthcare data through interactive charts, tables, and filters.



2. Objectives

The main objectives of this dashboard are:

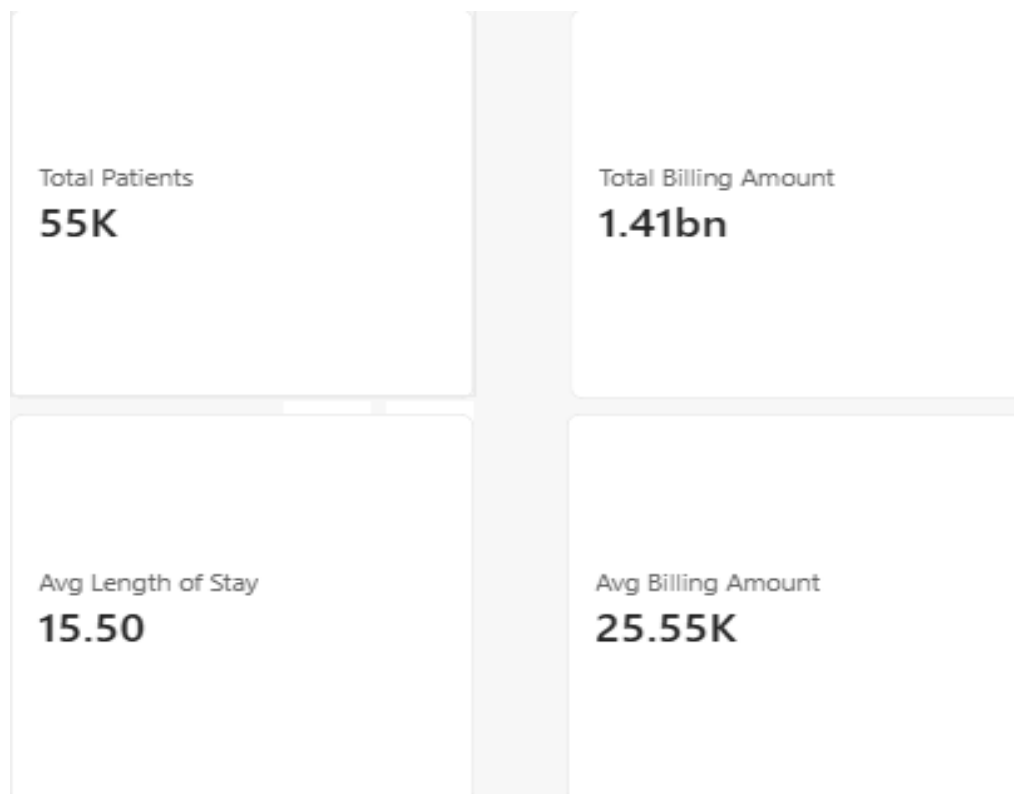
- To analyse the total number of patients.
- To analyse the total billing amount.
- To study billing amounts by insurance provider.
- To compare patients across different age groups.

- To analyse medical conditions and admission types.
- To understand the average length of patient stay.
- To provide an interactive healthcare report using Power BI.

3. Key Performance Indicators (KPI)

The dashboard contains four important KPI cards:

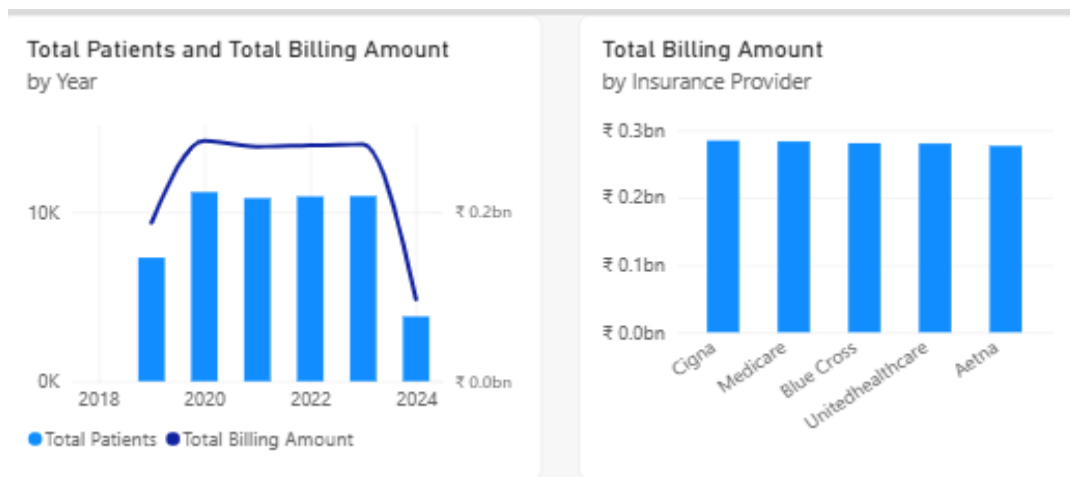
- **Total Patients – 55K**
- **Total Billing Amount – ₹1.41bn**
- **Average Length of Stay – 15.50**
- **Average Billing Amount – ₹25.55K**



These KPIs provide a quick summary of the overall healthcare data.

4. Total Patients and Total Billing Amount by Year

The combination chart shows the **total patients and total billing amount for different years**. The columns represent the number of patients, while the line represents the total billing amount.



This visual helps to understand the yearly changes in patient count and billing amount.

5. Total Billing Amount by Insurance Provider

The bar chart shows the total billing amount for different insurance providers such as:

- Aetna
- Blue Cross
- Cigna
- Medicare
- UnitedHealthcare

Insurance Provider

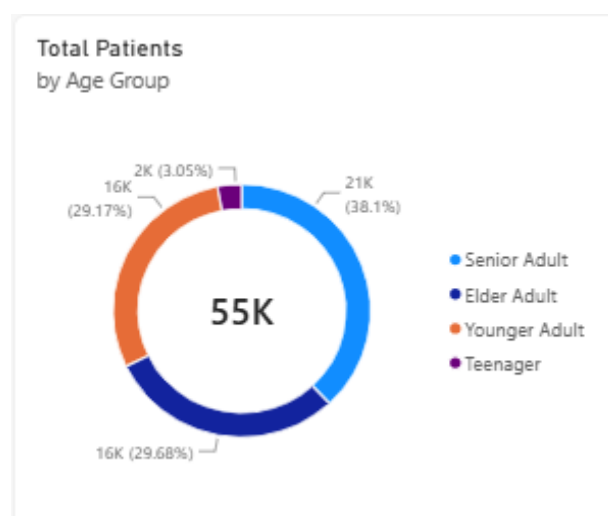
- ☐ Aetna
- ☐ Blue Cross
- ☐ Cigna
- ☐ Medicare
- ☐ Unitedhealthcare

This visual helps compare the billing amounts associated with different insurance providers.

6. Total Patients by Age Group

The doughnut chart represents the distribution of patients among different age groups:

- Senior Adult
- Elder Adult
- Younger Adult
- Teenager



The chart helps to understand how patients are distributed across different age groups.

7. Medical Condition and Admission Type

The stacked bar chart shows the relationship between **medical conditions and admission types**.

The medical conditions include:

- Arthritis
- Diabetes
- Hypertension
- Obesity
- Cancer
- Asthma

The admission types include:

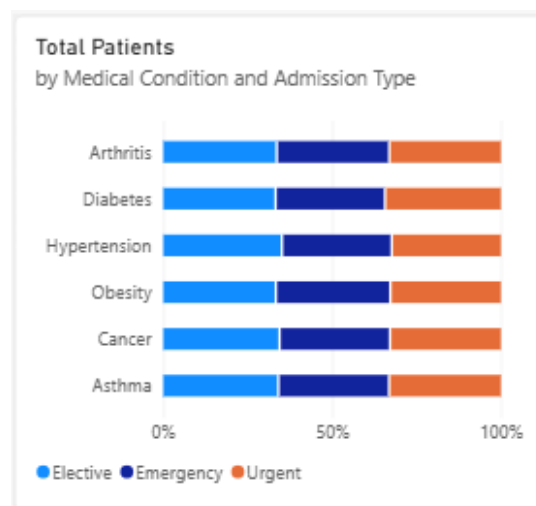
- Elective
- Emergency
- Urgent

Medical Condition	Arthritis		Asthma		Cancer	
Age Group	Total Patients	Avg Billing Amount	Total Patients	Avg Billing Amount	Total Patients	Avg Billing Amount
Elder Adult	2755	₹ 25,403.5417	2643	₹ 25,377.1204	2670	₹ 25,377.1204
Senior Adult	3486	₹ 25,727.4205	3512	₹ 25,486.3848	3507	₹ 25,486.3848
Teenager	296	₹ 25,638.0612	276	₹ 25,880.4484	282	₹ 25,880.4484
Younger Adult	2689	₹ 25,317.1022	2672	₹ 26,072.2072	2692	₹ 26,072.2072
Total	9226	₹ 25,508.2484	9103	₹ 25,638.5647	9151	₹ 25,638.5647

This visual helps compare admission types across different medical conditions.

8. Patient and Billing Analysis by Medical Condition and Age Group

The matrix visual provides detailed information about patient count and average billing amount based on **medical condition and age group**.



This table helps in comparing patient counts and billing amounts across different categories.

9. Interactive Filters

The dashboard contains interactive filters such as:

- Insurance Provider
- Admission Type
- Date of Admission

These filters allow users to select specific categories or dates and analyse the corresponding healthcare information.

Insurance Provider

- ☐ Aetna
- ☐ Blue Cross
- ☐ Cigna
- ☐ Medicare
- ☐ Unitedhealthcare

Admission Type

Elective	Urgent
Emergency	

Date of Admission

08-05-2019  07-05-2024 



10. Dashboard Features

The dashboard provides:

- Interactive KPI cards
- Charts and graphs
- Age group analysis
- Insurance provider analysis
- Medical condition analysis
- Admission type analysis
- Date filtering
- Detailed matrix analysis

11. Conclusion

The **Healthcare Analysis Dashboard** provides an interactive and easy-to-understand view of healthcare data. It helps analyse patients, billing amounts, medical conditions, admission types, age groups, and insurance providers. Power BI visuals and filters make it easier to explore the data and identify useful patterns.